

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)		$I = \frac{12.0\text{ V}}{450\ \Omega + 150\ \Omega} [= 0.020\text{ A}]$ or equivalent or $\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{150\ \Omega}{450\ \Omega + 150\ \Omega}$ or equivalent (1) Give mark if included in single equation (as below) $V_{\text{out}} = 0.020\text{ A} \times 150\ \Omega$ or $\frac{150\ \Omega}{450\ \Omega + 150\ \Omega} \times 12.0\text{ V}$ and = 3[.0 V] [as required] (1)		1	1	2	2	
	(b)		Any valid power calculation i.e. $\frac{3^2}{150}$ or $\frac{9^2}{450}$ or $\frac{12^2}{600}$ (1) For 2nd mark: - If used $\frac{9^2}{450}$ power in R_1 is 0.18 W which is < 0.5 W or which meets requirement - If used $\frac{12^2}{600}$ total power is 0.24 W and therefore power in R_1 is < 0.5 W (1) If $\frac{3^2}{150}$ don't award 2 nd mark ecf on current and V_{out} from part (a)		1	1	2	2	
	(c)		$\frac{1}{R} = \frac{1}{150} + \frac{1}{600}$ or equivalent or by implication (1) $R = 120\ [\Omega]$ (the modified R_2) (1) $V_{\text{out}} = 2.5\text{ [V]}$ or 2.53 [V] (1) So drop of > 0.4 V [therefore requirement not met] (give mark only if whole argument valid) (1) ecf on V_{out} from part (a)		1 1 1	1	4	2	
			Question 3 total	0	5	3	8	6	0